

Michael Phillips

mikeapstella@gmail.com | linkedin.com/in/michael-phillips-71101b28b/ | 617-991-5993

Education

Virginia Polytechnic Institute and State University

Bachelor of Science in Computer Science

Blacksburg, VA

August 2022 - May 2026

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C/C++, C#, HTML/CSS, Bash, PowerShell

Technologies: ReactJS, Redux, Azure, Dataverse, Power Apps, Microsoft 365, Git/Jira, OpenCV, OCR

Practices: Data Structures & Algorithms, Full-Stack Development, REST APIs, Data Engineering & Modeling, Storage Systems, Cloud Infrastructure, CI/CD, Agile/Scrum, Testing, ML, LLMs

Clearance: Active DoD Secret

Experience

Technomics, Inc.

Associate

Arlington, VA

July 2026 - Present

- Partner with DOE/NNSA stakeholders to translate ambiguous mission and operational requirements into client-facing software, structured data models, and analytical workflows.
- Design and build responsive Power Apps Canvas applications with dynamic filtering, querying, planning, reporting, and workflow features backed by Dataverse.
- Develop relational data models, ERDs, Dataverse schemas, and Microsoft 365 integrations that improve access to operational data and consistency across client processes.
- Build Python-based ingestion pipelines using OCR and document-processing libraries to extract, transform, validate, and load structured data from complex source documents.

The Math Tree

Co-Founder, Chief Technology Officer

Blacksburg, VA

May 2025 - Sept 2025

- Co-founded and served as Chief Technology Officer, owning software architecture, infrastructure, and technical execution for a full-stack web platform from concept through launch.
- Solely developed and maintained 5,000+ lines of front-end and back-end code, taking the platform beyond MVP in under 2 months while iterating rapidly on user and product feedback.
- Designed responsive ReactJS UI/UX, REST API endpoints, and relational database architecture, supporting growth from 10 to 200+ users within 2 weeks.
- Administered Microsoft Azure app services and database servers, maintaining scalable deployment architecture and reliable application availability.
- Led technical hiring, onboarding, and interviews while communicating engineering decisions to technical and non-technical stakeholders.

Peraton

Software Engineering Co-Op

Blacksburg, VA

May 2023 - December 2023

- Developed responsive ReactJS/CSS interfaces and scalable back-end functionality using SQL databases and custom REST API endpoints, translating client requirements into reliable full-stack features across deployment environments.
- Collaborated with engineers and clients in Agile development using Jira to scope requirements, test changes, resolve defects, and ship iterative releases against evolving technical needs.

Projects

Distributed Market Data and Signal Evaluation Platform

- Built Python pipelines and SQLite-backed storage to collect, normalize, persist, and analyze multi-timeframe market data using custom volatility, Bollinger Band, RSI, and risk indicators.
- Designed configurable concurrent worker execution for local LLM agents, aggregated probability-based outputs, and built a historical test harness to validate results against future price behavior.

Image Annotation and Computer Vision Tool

- Developed a ReactJS image annotation platform supporting image uploads, unlimited bounding boxes, XML export, and structured datasets for computer vision model training.
- Used Redux for predictable centralized state management and built reusable front-end components for high-volume annotation workflows.

ASL Hand Sign Recognition Model

- Developed a transfer-learning and OpenCV pipeline to classify American Sign Language hand-signed letters from video input using custom labeled datasets.
- Implemented preprocessing, augmentation, evaluation, and inference workflows to improve recognition consistency and support repeatable model experimentation.